Building Neural Networks in Python

A Practical Introduction using Keras

Workshop Facilitator

April 23, 2026

Workshop Agenda

- **Part 1: The Theory (20 mins)**
  - How Neural Networks learn
  - The Keras API
  - The 5-Step Deep Learning Workflow
- **Part 2: Hands-On Coding (60 mins)**
  - Environment setup
  - Data preparation & shapes
  - Building, training, and tuning a model

What is a Neural Network?

- A machine learning algorithm inspired by the human brain.
- Designed to recognize patterns in data (numbers, text, images).
- Instead of hard-coding rules (e.g., if $X > 5$ then $Y = 1$), we provide data and answers, and the network "learns" the rules.
- This is done using *training data* and *training labels*

Network Architecture: Deep, Shallow, Wide, and Narrow

We often hear these terms used to describe a model's **architecture** sometimes referred to as **topology** in more theoretical applications.

Depth (Number of Layers)

- **Shallow:** 1 to 2 hidden layers. Best for simpler data. Faster to train and less likely to overfit.
- **Deep:** 3+ hidden layers (This is "Deep Learning"). Best for hierarchical data where patterns build on each other (e.g., combining edges into shapes, and shapes into faces).

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- **Narrow:** Few nodes (e.g., 8, 16). Forces the network to compress information and only learn the most essential, defining features.
- **Wide:** Many nodes (e.g., 128, 512). Gives the model a massive capacity to capture diverse, complex patterns, but demands much more data to train effectively.

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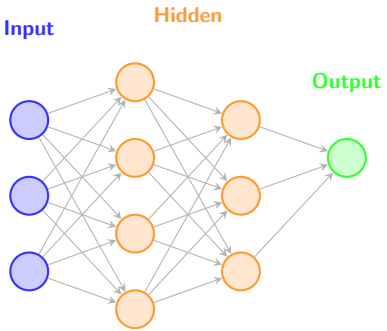
Rule of Thumb: *Start shallow and narrow. Only add depth and width when your model proves it isn't complex enough to learn the data!*

Anatomy of a Feedforward Network

- **Input Layer:** One node for each feature in your dataset.
- **Hidden Layers:** Where the mathematical transformations occur. "Deep" learning implies multiple hidden layers.
- **Output Layer:** The final prediction (e.g., a single probability for binary classification).



Data flows strictly from left to right during prediction.

Weights & Biases

How does the network actually learn?

- Every connection between nodes has a **Weight** (importance).
- Every node has a **Bias** (an offset).
- **The Goal of Training:** Find the exact combination of weights and biases that minimizes prediction error.
- It does this through an iterative process called **Gradient Descent**.

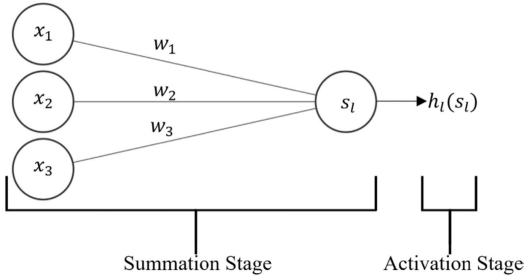


FIGURE A2 Neural network summation and activation stages.

$$s_l = \sum_{j=1}^{J} \left(x_j \times w_j \right) + b_l \qquad \text{(A4)}$$

Quan, Y., & Wang, C. (2026). Using multilabel classification neural network to detect intersectional DIF with small sample sizes. British Journal of Mathematical and Statistical Psychology, 00, 1–38. https://doi.org/10.1111/bmsp.70041

Adding Non-Linearity: Activation Functions

Without activation functions, a neural network is just a linear regression model. (This is actually the default in Keras)

- **ReLU (Rectified Linear Unit):**
  - If input is negative, output 0. If positive, output the input.
  - The default choice for *Hidden Layers*.
- **Sigmoid:**
  - Squeezes any number into a probability between 0 and 1.
  - The default choice for the *Output Layer* in binary classification.
- **Linear:**
  - No activation is applied, output is simply $f(x) = x$
  - The default choice for the *Output Layer* in regression.

Classification vs. Regression in Keras

The strength of Keras is that changing the nature of your model often just requires changing the **Output Layer** and the **Loss Function**.

Binary Classification

- *Goal:* Predict a category (e.g., Pass/Fail).
- *Output Nodes:* 1
- *Activation:* `sigmoid` (Probabilities)
- *Loss:* `binary_crossentropy`

Regression

- *Goal:* Predict a continuous number (e.g., Final Score).
- *Output Nodes:* 1
- *Activation:* `linear` (or omit)
- *Loss:* `mse` (Mean Squared Error)

Why Keras?

- **High-Level API:** Designed for humans. It translates your intent into complex math.
- **TensorFlow Engine:** Runs on top of Google's TensorFlow for production-grade speed.
- **Lego Blocks:** Building a model is just stacking layers like blocks.

The Keras Workflow

1. **Prepare Data**
2. **Define Architecture**
3. **Compile Model**
4. **Fit Model**
5. **Evaluate & Predict**

Step 1: Data Preparation

Neural networks are brittle. They require strict data formats:

- **Numbers Only:** All categorical text must be encoded.
- **Scaling is Recommended:** Features should generally be scaled between 0 and 1, or standardized (mean 0, variance 1). Large unscaled numbers produce unstable gradients.
- **Train/Test Split:** We must hold out data the model has never seen to test if it actually learned, or just memorized.

Step 1.5: Understanding Shapes

The most common error beginners face in Keras is a *shape mismatch*.

- Data is fed in batches.
- If you have 100 rows and 5 features, your data shape is $(100, 5)$.
- The **Input Layer** must be explicitly told to expect 5 features using `input_shape=(5,)`.
- If you train a network using a data shape $(100, 5)$ and then try to fit it to data with shape $(100, 10)$ the model will crash and the error is not very helpful. The column count (or feature length) is important.

Example Error:

```
ValueError: Input 0 of layer "dense" is incompatible with the layer:
    expected axis -1 of input shape to have value 5, but received input with
     shape (None, 10)

```

Step 2: Defining the Architecture

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Initialize an empty sequential model
model = Sequential()

# Add a hidden layer (must specify input_shape on the first layer)
model.add(Dense(64, activation='relu', input_shape=(5,)))

# Add another hidden layer
model.add(Dense(32, activation='relu'))

# Output layer: 1 node, sigmoid for binary probability
model.add(Dense(1, activation='sigmoid'))

```

Step 3: Compiling the Model

We must define the rules for training before we begin.

```
model.compile(
    optimizer='adam',              # The algorithm updating the weights
    loss='binary_crossentropy',    # How we calculate error
    metrics=['accuracy']           # Human-readable performance tracking
)

```

Note: `adam` is a default optimizer that dynamically adjusts its learning rate.

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Loss is the mathematical penalty the optimizer uses to update weights.
Metrics are human-readable scores to help you monitor performance.

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- **Loss:** `binary_crossentropy`
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- **Loss:** `categorical_crossentropy`
  - For 3+ categories (e.g., predicting Letter Grades A, B, C, D, F).
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  - Squares the errors, which heavily penalizes massive outliers. Good for gradient descent.
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Note: *Always monitor the* **validation** *metrics (e.g.,* `val_loss`*) to detect overfitting! Many other loss function and metrics exist.* **You can also write your own!**

Step 4: Training (Fitting) the Model

```
history = model.fit(
    X_train,
    y_train,
    epochs=50,           # Full passes through the dataset
    batch_size=32,       # Rows processed before updating weights
    validation_split=0.2 # Automatically hold out 20% for testing
)

```

Note:

1. As you increase the number of epoch the network *usually* becomes more accurate, but this will add computational complexity.
2. Increase or decreasing the batch size can help the model learn more complex patterns.
3. The default validation split is naive. It will just split the training data into chunks. Writing code to perform specific splitting rules is often better.

What happens during `model.fit()`?

Phase 1: Training (Learning the Data)

1. **Forward Pass:** Training batches go in, predictions come out.
2. **Loss Calculation:** Compare predictions to the actual answers.
3. **Backpropagation:** Calculate how much each weight contributed to the error.
4. **Update:** The Optimizer adjusts the weights to reduce the error.

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Phase 2: Validation (The Reality Check)

- At the end of every epoch, the model evaluates the held-out **Validation Data**.
- The model is fit to the unseen data and the predicted values are compared to the ground truth labels.
- The model calculates loss and metrics, but **does not update its weights**.
- This acts as an independent check to ensure the model is learning general patterns, rather than just memorizing the training set.

Step 5: Evaluation & Prediction

```
# Evaluate on the final, unseen test set
loss, accuracy = model.evaluate(X_test, y_test)
print(f"Final Test Accuracy: {accuracy:.4f}")

# Generate predictions for brand new data
new_probabilities = model.predict(X_new_data)

# Convert probabilities to class labels (0 or 1)
new_classes = (new_probabilities > 0.5).astype("int32")

```

- `X_text` is the held out testing data observations.
- `y_test` is the ground truth for the testing data.
- `X_new_data` is the new data we collect without labels.

Monitoring Training: Overfitting

During the hands-on session, watch your `val_loss` (Validation Loss).

- If Training Loss goes down, but Validation Loss goes up...
- Your model is **Overfitting**. It is memorizing the training data and failing to generalize.
- *Solution:* Stop training earlier, use a simpler model (fewer nodes), or get more data. Keras has a helpful function called `EarlyStopping` and `callback` to help.

Beyond Feedforward: Specialized Architectures

So far, we have built **Feedforward (Dense)** networks, where data flows straight from input to output. For more advanced models, Keras has built-in layers for specialized data structures:

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- **Convolutional Neural Networks (CNNs)**
  - *The Concept:* They use sliding mathematical filters to detect spatial patterns like edges, shapes, and textures.
  - *Best for:* Grid-like data such as images, computer vision tasks, or scanning documents.
  - *Keras Layer:* `Conv2D`

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  - *Best for:* Grid-like data such as images, computer vision tasks, or scanning documents.
  - *Keras Layer:* `Conv2D`
- **Recurrent Neural Networks (RNNs & LSTMs)**
  - *The Concept:* They process data one step at a time, maintaining an internal state or "memory" of previous inputs to understand context.
  - *Best for:* Sequential data like time-series tracking, natural language processing, or chronological sequences of student test responses.
  - *Keras Layer:* `LSTM` or `SimpleRNN`

Callbacks (Smart Training)

Callbacks are functions that Keras runs at the end of every epoch to monitor training and take action automatically.

- **EarlyStopping:** Halts training the moment validation loss stops improving. You can specify the convergence criteria (Defaults min_delta = 0 zero) and the value to monitor (Default: 'val_loss')
- **ModelCheckpoint:** Automatically saves the weights of your *best* epoch, rather than your *last* epoch.

```
from tensorflow.keras.callbacks import EarlyStopping

# Stop training if val_loss doesn't improve for 5 epochs in a row
early_stop = EarlyStopping(monitor='val_loss', patience=5,
            min_delta = 0.001)

history = model.fit(X_train, y_train, epochs=100,
                    callbacks=[early_stop]) # Pass to the fit function

```

The Functional API

So far, we used the `Sequential` model (a straight line). But real-world research often requires complex, branching architectures (e.g., combining student test responses with demographic features to detect assessment bias).

The **Functional API** allows you to build models with multiple inputs, multiple outputs, or shared layers.

```
from tensorflow.keras.layers import Input, Dense,
    Concatenate
from tensorflow.keras.models import Model

# Two separate data streams
test_responses = Input(shape=(50,))
demographics = Input(shape=(5,))

# Process streams independently
branch_A = Dense(32, activation='relu')(test_responses)
branch_B = Dense(8, activation='relu')(demographics)

# Merge them together for a final prediction
merged = Concatenate()([branch_A, branch_B])
output = Dense(1, activation='sigmoid')(merged)

model = Model(inputs=[test_responses, demographics],
    outputs=output)
```

- Layers are treated like functions: `Layer()(previous_layer)`
- Essential for modern machine learning research.
- Highly flexible, yet uses the exact same `compile()` and `fit()` workflow!

Automating the Search with KerasTuner

Manually guessing the best number of nodes or layers is exhausting. **KerasTuner** automates the trial-and-error process by training dozens of models and keeping the best one.

```
import keras_tuner as kt
from tensorflow.keras.layers import Input, Dense
from tensorflow.keras.models import Sequential

# 1. Define a model-building function with "hp" (hyperparameters)
def build_model(hp):
    model = Sequential()
    model.add(Input(shape=(3,)))

    # Let the tuner choose the number of nodes (between 8 and 32)
    hp_units = hp.Int('units', min_value=8, max_value=32, step=8)
    model.add(Dense(hp_units, activation='relu'))

    model.add(Dense(1, activation='sigmoid'))
    model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
    return model

```

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```
1
2 # 2. Instantiate the Tuner (Random Search, Hyperband, etc.)
3 tuner = kt.RandomSearch(
4     build_model,
5     objective='val_accuracy',
6     max_trials=5 # Try 5 different combinations
7 )
8
9 # 3. Start the search!
10 tuner.search(X_train_scaled, y_train, epochs=10, validation_split=0.2)
11
12
```

Note: This is not always the best way to tune. This is very computationally expensive method. Sometimes you can review prior research to inform the tuning and you can specify a more fine-tuned grid.

Manual Tuning

We can also manually tune (which is usually what I do). You can create a search grid and write keras and python code that iterates over the grid and save any metrics you want. We can do this using JSON files

```
1
2  #Three Group Training JSON
3  {
4      "groups": 3,
5      "architectures":              ["Single_Funnel",  "Contextual"],
6      "split_depths":               [0, 1, 2, 3],
7      "use_within_features": [true, false],
8      "trunk_configs":           [ [32, 16], [64, 32], [128, 64], [128, 64, 32], [64,
       32, 16]],
9      "dropout_rates":           [0.1, 0.2],
10     "learning_rates":          [0.0001, 0.001],
11     "batch_sizes":              [64, 128, 256],
12     "loss_weight_a":         [1.0, 1.25, 1.5, 2.0]
13 }
```

Note: This results in 3,840 configurations to test. While this is very large, the grid is targeted for values I have identified as important. This takes about 2 weeks to run.

Let's Code!

Time to open your notebooks.

We will now work through a complete end-to-end classification and regression example together.

Please navigate to the provided Colab link to get started.